MolSanity: Auditing the Reliability of Feature Attributions for Molecular GNNs under Scaffold Shift

MolSanity contributors

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https://github.com/Kar488/molsanity

Abstract. Feature attributions are routinely used to explain which atoms drive a molecular graph neural network's prediction, yet they are seldom validated against what the molecule actually contains. The dominant way to check an attribution is *faithfulness* — does the explanation track the model? — but faithfulness says nothing about *correctness* — does the explanation recover the true causal substructure? We present MolSanity, a reliability audit that measures both on the same molecules, together with coherence, cross-checkpoint stability and calibration linkage, and stratifies every metric by confidence/correctness regime under Bemis–Murcko scaffold shift. Across 81 audited cells spanning 13 datasets, 5 backbones and 6 attributors, we find that faithfulness and correctness are essentially *uncorrelated* (Spearman $\rho = -0.03$, $p = 0.88$, $n = 29$ ground-truth-bearing cells). On MUTAG the four gradient-family attributors span only 0.073 in occlusion faithfulness while their ground-truth AUROC spans 0.514 (0.026–0.540, chance $= 0.5$): faithfulness barely discriminates between methods whose correctness differs by more than half an AUROC. This dissociation is regime-dependent. On a synthetic task with *exact* node ground truth and $n = 120$ evaluation molecules, ranking attributors by faithfulness recovers the ground-truth-best method ($\rho = +0.94$, $p = 0.005$); under scaffold shift the same ranking inverts in sign, and the field-standard Fidelity+ and GraphFramEx characterisation scores select attributors whose ground-truth AUROC is 0.112 and 0.026 against a best of 0.540 (paired Wilcoxon $p < 0.001$). Reliability also varies by backbone: on exact ground truth, localisation orders GCN > AttentiveFP > MPNN > GINE > GAT, and the *most accurate* model (GINE, test accuracy 1.00) is not the best-explained. We report negatives plainly, including degenerate cells and a directionality caveat that makes the regression faithfulness numbers weaker evidence than they appear. MolSanity is open source, with 35 pretrained checkpoints committed so every table reproduces without retraining.

1 Introduction

Graph neural networks (GNNs) are now standard for molecular property prediction (MPP), and post-hoc feature attribution is the standard way practitioners ask *why* a model made a call. A medicinal chemist looking at a toxicity prediction wants to know which substructure drove it. Integrated Gradients [24], GNNExplainer [30] and PGExplainer [18] all return such a map, and all of them return one whether or not the map means anything.

The prevailing way to check that a map means something is *faithfulness*: does the explanation track the model's own computation? Fidelity± and related perturbation scores [32], and evaluation frameworks built on them [3, 2, 17], all operationalise this idea. Faithfulness is necessary — an explanation unmoored from the model is useless — but it is not sufficient. A model can key on a spurious correlate, and an attribution can report that correlate perfectly. The explanation is then maximally faithful and completely wrong about the chemistry. Sanity checks in vision found exactly this class of failure [1], and in molecules the stakes are concrete: the attribution is read as a structure–activity claim.

Distribution shift sharpens the problem. MPP models are deployed on chemistry unlike their training set, which is why Bemis–Murcko scaffold splits [4] are the accepted evaluation protocol [26, 29]. Attribution reliability under that shift is, however, rarely measured at all.

A concrete failure. Take MUTAG with a GINE backbone under a scaffold split (Table 2). Saliency attains occlusion faithfulness $\rho = 0.376$ and Fidelity+ $= 0.267$ — respectable by any faithfulness-first standard, and the *best* characterisation score of the six attributors we test. Its ground-truth AUROC against the nitro-motif mask is 0.026, where 0.5 is chance: it is strongly anti-aligned with the substructure the label is built on. Integrated Gradients, at a very similar faithfulness ($\rho = 0.414$, Fidelity+ $= 0.252$), reaches 0.540. A practitioner ranking by the field-standard scores would adopt the attributor that is confidently pointing at the wrong atoms.

Contributions.

1. **A reliability audit, not another attributor.** MolSanity composes occlusion faithfulness, ground-truth localisation, an atom- and motif-level coher-

ence battery, cross-checkpoint stability and calibration linkage, and stratifies all of them by confidence/correctness regime under scaffold shift (§3). It wraps canonical implementations [15, 6, 16] rather than reimplementing them.

2. **Evidence that faithfulness does not imply correctness.** Over 29 ground-truth-bearing cells the two are uncorrelated ($\rho = -0.03$, $p = 0.88$); on MUTAG a 0.073 spread in faithfulness hides a 0.514 spread in correctness (§5.1).

3. **A regime-dependent selection failure, tested for significance.** With exact ground truth in-distribution, faithfulness-based ranking picks the correct attributor ($\rho = +0.94$, $p = 0.005$); under scaffold shift Fidelity+ and characterisation mis-select, with per-molecule paired Wilcoxon $p < 0.001$ (§5.2).

4. **Backbone- and task-dependence, with negatives kept in.** Localisation on exact ground truth orders GCN > AttentiveFP > MPNN > GINE > GAT, and the most accurate model is not the best explained (§5.3); we report degenerate cells and a directionality caveat on the regression numbers rather than suppressing them (§5.5, §7).

2 Related work

Attribution methods. Gradient-family attributions — Saliency [22], Input×Gradient [21], Guided Backpropagation [23] and Integrated Gradients [24] — transfer directly to graphs through Captum [15]. Graph-specific methods learn an explanatory mask: GNNExplainer optimises one per instance [30], PGExplainer amortises a parametric mask generator across instances [18], and SubgraphX searches connected subgraphs with Shapley scoring [31]. We audit these methods; we do not propose a new one.

Evaluating explanations. Fidelity± and sparsity are the field-standard perturbation metrics [32]. GraphFramEx [3] systematises them and adds a characterisation score combining Fidelity+ and Fidelity−; GraphXAI [2] supplies synthetic graphs with ground-truth explanations and a metric suite; DIG [17] packages methods and metrics as a library. In vision, *Sanity Checks* [1] showed some saliency maps are invariant to randomising the model, and ROAR [9] showed retraining-based evaluation can reverse method rankings; faithfulness has also been argued to need careful definition [12]. Our contribution is orthogonal to all of these: we measure faithfulness *and* ground-truth correctness on the same molecules and show the two dissociate as a function of distribution shift.

Molecular attribution. Sanchez-Lengeling et al. [20] benchmark GNN attributions on synthetic molecular tasks with known ground truth, and remains the closest prior work; McCloskey et al. [19] use attribution to recover binding mechanism, and Jiménez-Luna et al. [13] survey explainable AI for drug discovery. Relative to these, MolSanity adds the shift axis (scaffold splits with regime stratification), a motif-native audit built on RDKit ring/Murcko/BRICS decompositions, stability across training checkpoints, calibration linkage, and a head-to-head test of whether faithfulness-based model selection agrees with ground truth.

Positioning. GraphFramEx, GraphXAI and DIG answer "is this explanation faithful, and how does it score against a synthetic ground truth in distribution?" MolSanity answers "*when does faithfulness stop predicting correctness*, and for which backbone, attributor and chemistry?" Table 3 shows this is not a hypothetical distinction: the GraphFramEx characterisation score selects the correct attributor in-distribution and the wrong one under shift, on the same benchmark, with the same code.

3 The MolSanity audit

Notation. A molecule is a graph $G = (V, E)$ with $n = |V|$ atoms, node features $X \in \mathbb{R}^{n\times d}$ and edge features. A trained model f maps G to logits $f(G) \in \mathbb{R}^C$ (classification) or a scalar (regression). An attributor $\mathcal{A}$ returns node scores $a = \mathcal{A}(f, G) \in \mathbb{R}^n_{\geq 0}$; we use unsigned magnitudes throughout and min–max normalise to $[0, 1]$ where a common scale is needed. RDKit gives a motif decomposition $\mathcal{M} = \{M_1, \dots, M_K\}$, $M_k \subseteq V$, from smallest-set-of-smallest-rings, Bemis–Murcko scaffolds and BRICS fragments. The motif-level score is $s_k = \sum_{i\in M_k} a_i$.

Occlusion faithfulness. For each motif we mask its atoms and record the drop in the explained-class logit,

$$\Delta_k \;=\; f_c(G) - f_c(G \setminus M_k)\,,$$

and report $\rho_{\text{occ}} = \text{Spearman}(s_{1:K}, \Delta_{1:K})$. **Higher is more faithful**: the attribution ranks motifs the same way occlusion does. $\rho_{\text{occ}} < 0$ means the attribution ranks motifs in the *opposite* order to their causal effect on the model. We also report top-1 agreement, and the field-standard Fidelity+ (probability drop when the salient atoms, top quintile, are removed; higher is better), Fidelity− (the drop when non-salient atoms are removed; lower is better), sparsity, and the GraphFramEx characterisation score, the weighted harmonic mean of Fidelity+ and 1−Fidelity− [3].

Ground-truth correctness. Where a per-atom ground-truth mask $g \in \{0, 1\}^n$ exists, correctness is the AUROC of the normalised attribution against g, with AUPRC alongside. Chance is 0.5; *below* 0.5 means anti-alignment. AUROC is undefined for a degenerate mask and we emit `--` rather than a number. Two sources of g are used, and we keep them distinct throughout: SynthMotifs/SynthMotifsXL, synthetic Barabási–Albert graphs with an injected house or 5-cycle motif, give *exact* node labels; MUTAG gives a chemically motivated nitro-group *proxy*, since it ships no annotator labels [5].

Coherence. Independent of ground truth, an attribution should be concentrated and chemically contiguous. We report the Gini coefficient of a, the mass in the top 20% of atoms, the fraction of salient atoms lying in a single connected component, and the share of attribution mass in the top motif.

Cross-checkpoint stability. Training saves an intermediate checkpoint at one fifth of the epoch budget. Stability is Spearman$(s^{\text{early}}, s^{\text{final}})$ over motifs for the same molecule: an explanation that reorders the chemistry as training proceeds is reporting optimisation noise as much as structure.

Calibration linkage and regimes. Logits are temperature-scaled on validation [8]. Per molecule we take the confidence $\hat{p} = \max_c \operatorname{softmax}(f(G)/T)_c$ and assign a regime: *confident correct* ($\hat{p} \geq \tau$, correct), *confident error* ($\hat{p} \geq \tau$, wrong) — the dangerous quadrant — and *borderline* ($\hat{p} < \tau$), with $\tau = 0.8$. Calibration linkage is the Spearman correlation between $1 - |\hat{p} - \mathbf{1}[\text{correct}]|$ and a reliability metric, testing whether better-calibrated predictions carry more trustworthy attributions. Every metric above is computed per molecule and aggregated with bootstrap confidence intervals and paired Wilcoxon tests on shared molecules.

4 Experimental setup

Grid. The audit unit is a *cell*: one (dataset $\times$ backbone $\times$ attributor $\times$ split). We report 81 cells — 64 classification and 17 regression — over 13 datasets, 5 backbones and 6 attributors (Table 1). The grid is deliberately incomplete: it is not a full Cartesian product, because attributors were added on top of already-trained backbones and a few combinations are unsupported (PGExplainer's parametric mask is trained against class logits and does not apply to regression).

Datasets. Tier 1 provides ground truth: SynthMotifs (200 graphs) and SynthMotifsXL (600 graphs), generated offline with exact node labels, and MUTAG (188 molecules) with the nitro proxy. Tier 2 and 3 are real MPP tasks with no per-atom ground truth: BBBP, BACE, ClinTox, SIDER and Tox21 from MoleculeNet [26]; DILI and hERG from the Therapeutics Data Commons [11]; and ESOL, FreeSolv and Lipophilicity as regression [26]. Multi-task panels are audited as single-task binary views (Tox21 NR-AR, SIDER task 0, ClinTox CT_TOX). TDC molecules are featurised with the same SMILES encoder as MoleculeNet so the backbones are drop-in.

Table 1: The audited grid. n_{mol} is the number of evaluation molecules per cell (capped by the configuration budget). Only the three Tier-1 datasets carry per-atom ground truth; for the rest, correctness is undefined and reported as -- rather than imputed.

Dataset	Task	Cells	n_{mol}	Node ground truth
SynthMotifs	clf.	11	20	exact
SynthMotifsXL	clf.	6	120	exact
MUTAG	clf.	12	20	proxy
BBBP	clf.	8	30	none
BACE	clf.	5	30	none
ClinTox	clf.	4	30	none
SIDER	clf.	5	30	none
Tox21	clf.	4	30	none
DILI	clf.	4	30	none
hERG	clf.	5	30	none
ESOL	reg.	7	20	none
FreeSolv	reg.	5	20	none
Lipophilicity	reg.	5	20	none
Total		**81**		

Backbones and attributors. GINE [28, 10], GCN [14], GAT [25], MPNN [7] and AttentiveFP [27], all via PyTorch Geometric [6]. Attributors: Integrated Gradients, Saliency, Input$\times$Gradient and Guided Backpropagation through Captum [15], plus GNNExplainer and PGExplainer.

Splits and budget. The primary protocol is a Bemis–Murcko scaffold split [4] (80/10/10), made class-aware for imbalanced sets by moving whole scaffold groups so that no fold is single-class — no scaffold is split across folds. Random splits are included as an in-distribution reference. Results come from three CPU-tractable configurations: a broad matrix (30 epochs, 15 IG steps, 20 audited molecules per cell), a Tier-2/3 configuration (60 epochs, 25 IG steps, 30 molecules, inverse-frequency class weighting and AUC-based model selection), and a ground-truth benchmark configuration (40 epochs, 25 IG steps, the whole 120-molecule test fold of SynthMotifsXL). **This is a reduced budget**, chosen so the audit runs on CPU; it bounds the precision of every per-cell number and we treat it as a first-class limitation (§7). Seeds, library versions and hardware are logged per run; stochastic explainers are seeded per molecule so an explanation depends only on (molecule, method). 35 trained checkpoints are committed, so the tables regenerate without retraining.

5 Results

5.1 Faithfulness does not imply correctness

Figure 1 plots every ground-truth-bearing cell in the faithfulness–correctness plane. Across all 29 such cells

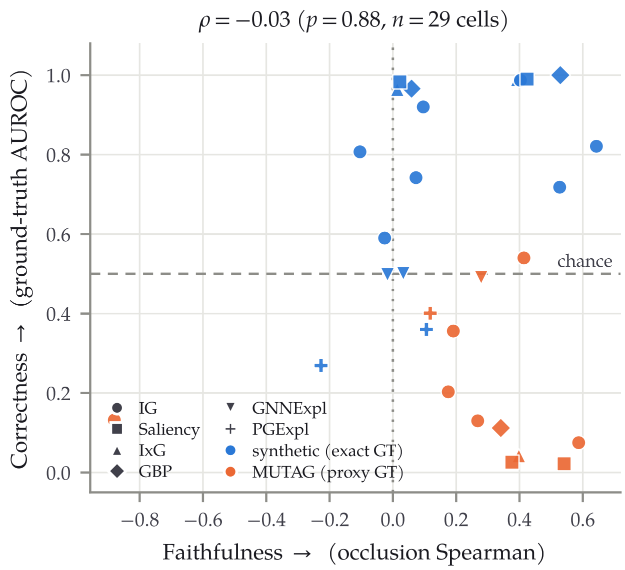

Figure 1: Faithfulness carries almost no information about correctness. Each point is one audited cell with per-atom ground truth ($n = 29$); x is occlusion Spearman (higher = more faithful), y is ground-truth AUROC (higher = more correct, 0.5 = chance). Marker encodes attributor, colour encodes ground-truth source. Points in the lower right are faithful and wrong.

the two axes are statistically unrelated: Spearman $\rho = -0.03$ ($p = 0.88$). Knowing how faithful an attribution is tells you essentially nothing about whether it recovers the right substructure.

The mechanism is visible in Table 2. On MUTAG × GINE under scaffold shift, the four gradient-family attributors occupy a narrow faithfulness band — ρ_{occ} from 0.341 to 0.414, a spread of 0.073 — while their ground-truth AUROC runs from 0.026 to 0.540, a spread of 0.514. Saliency and Input×Gradient sit near 0.03–0.04, far *below* chance, i.e. systematically anti-aligned with the nitro motif, yet they are among the most faithful and score the highest Fidelity+ and characterisation values in the table. Integrated Gradients and GNNExplainer recover the motif far better at comparable faithfulness.

We are precise about what this does and does not show. Ranking these six attributors by *our* occlusion ρ happens to put Integrated Gradients on top, which is also the ground-truth-best method — the argmax agrees. What fails is the *relationship*: faithfulness is nearly constant across methods whose correctness differs by more than half an AUROC, so it cannot discriminate between them, and the rank correlation over the six is -0.03 (Table 3). The field-standard metrics are worse than uninformative here: Fidelity+ ranks Guided Backpropagation first and characterisation ranks Saliency first, with ground-truth AUROC of 0.112 and 0.026 respectively.

5.2 The dissociation is regime-dependent

Faithfulness is not always uninformative — and saying so is part of the result. Table 3 and Figure 2 contrast

Table 2: The faithful-but-wrong case. MUTAG × GINE, scaffold split, 20 evaluation molecules, sorted by correctness. Occlusion ρ and Fidelity+ (higher = more faithful) barely separate the four gradient methods, whose ground-truth AUROC spans 0.514. Ground truth here is the nitro *proxy*, not annotator labels.

Attributor	GT AUROC	Occ. ρ	Fid+	Fid−	Motif top1
Integrated Grad.	0.540	0.414	0.252	0.245	0.979
GNNExplainer	0.491	0.279	0.113	0.278	0.987
PGExplainer	0.401	0.118	0.026	0.294	1.000
Guided BP	0.112	0.341	0.272	0.229	0.990
Input×Grad.	0.042	0.398	0.251	0.225	0.998
Saliency	0.026	0.376	0.267	0.230	0.997

two regimes. In distribution, on SynthMotifsXL with exact ground truth and the full 120-molecule test fold, faithfulness tracks correctness well: rank correlations are $+0.94$ ($p = 0.005$) for occlusion ρ, $+0.83$ ($p = 0.042$) for Fidelity+ and $+0.77$ ($p = 0.072$) for characterisation, and all three selectors pick the ground-truth-best attributor. A faithfulness-only benchmark is adequate in this regime.

Under scaffold shift on MUTAG the same three correlations invert in sign (-0.03, -0.37, -0.49), and Fidelity+ and characterisation actively mis-select: they top-rank attributors whose ground-truth AUROC is 0.112 and 0.026 against a best of 0.540. Comparing the mis-selected attributor with the ground-truth-best on the *same* molecules, the per-molecule paired Wilcoxon test rejects at $p < 0.001$ for both.

The honest reading of the significance is mixed and we state it plainly. The *selection error* is well supported: it is a paired test over 20 molecules with $p < 10^{-4}$. The *rank correlations* on the shift arm are not individually significant — they are computed over only six attributors, so $p = 0.957$, 0.468 and 0.329 — and we therefore report the sign reversal as a consistent pattern across three independent selectors rather than as an individually significant statistic. The in-distribution correlations, over the same six attributors, do reach significance for two of the three selectors.

5.3 Reliability depends on the backbone

Holding the attributor fixed at Integrated Gradients and varying the backbone on SynthMotifs, where ground truth is exact, localisation orders GCN (0.920) > AttentiveFP (0.821) > MPNN (0.807) > GINE (0.742) > GAT (0.718) (Figure 3, Table 4). The spread is 0.202 AUROC from the same attributor on the same task — comparable to the spread between attributors — so "which explainer" and "which architecture" are of similar importance to reliability.

Predictive accuracy does not buy explanation quality. GINE reaches test accuracy 1.00 and AUC 1.00 on this task, and is fourth of five on localisation; GCN, at accuracy 0.90, localises best. AttentiveFP is the weakest classifier here (accuracy 0.60) yet is second on localisation. Model selection by predictive performance there-

Table 3: Would a faithfulness-only evaluation pick the right attributor? ρ_{rank} is the Spearman correlation between the selector and ground-truth AUROC across the six attributors (so $n = 6$; these correlations are underpowered and only two reach $p < 0.05$). "Mis-selection" compares the selector's top pick against the ground-truth-best attributor on the same molecules with a paired Wilcoxon test. In distribution every selector is correct; under scaffold shift the two field-standard selectors are not.

Regime	Selector	ρ_{rank}	p	Top pick (GT AUROC)	Mis-selection
in-distribution (SynthMotifsXL, n =120)	Occlusion ρ (ours)	+0.94	0.005	GuidedBP (1.000)	no
	Fidelity+	+0.83	0.042	GuidedBP (1.000)	no
	Characterisation	+0.77	0.072	GuidedBP (1.000)	no
scaffold shift (MUTAG, n =20)	Occlusion ρ (ours)	-0.03	0.957	IG (0.540)	no
	Fidelity+	-0.37	0.468	GuidedBP (0.112)	**yes**, $p < 0.001$
	Characterisation	-0.49	0.329	Saliency (0.026)	**yes**, $p < 0.001$

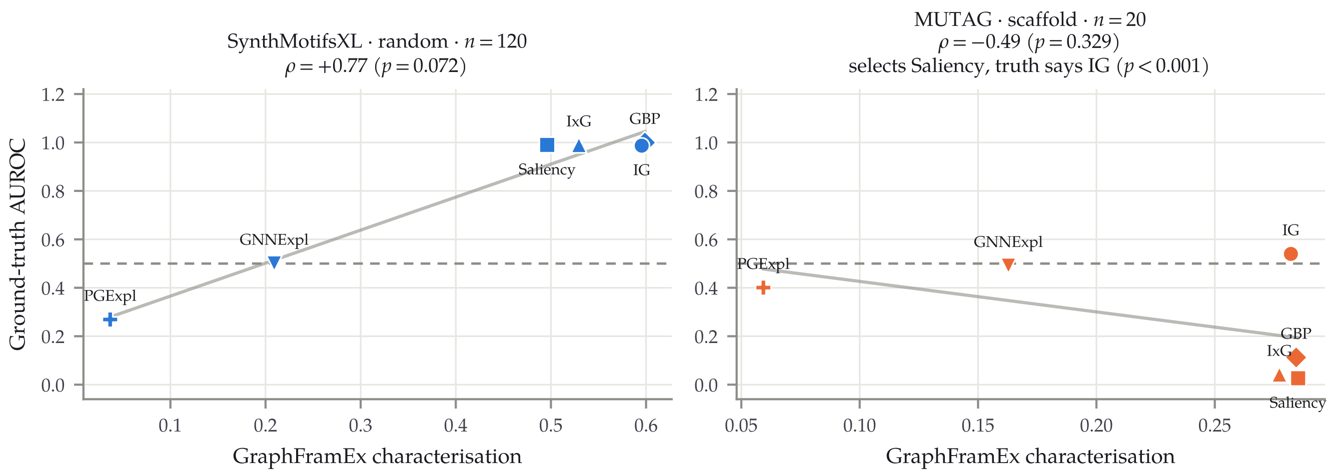

Figure 2: The same selector, two regimes. GraphFramEx characterisation (x) against ground-truth AUROC (y), one point per attributor. Left: in-distribution with exact ground truth, the relationship is positive and the top-ranked attributor is the correct one. Right: under scaffold shift the relationship inverts and the selector top-ranks Saliency, whose ground-truth AUROC is 0.026 against Integrated Gradients' 0.540. ρ values are rank correlations over the six attributors; the mis-selection p-value is a per-molecule paired Wilcoxon test.

Table 4: Backbone comparison on exact ground truth (SynthMotifs, Integrated Gradients, scaffold split, 20 evaluation molecules). Predictive performance and attribution correctness are not aligned.

Backbone	GT AUROC	Occ. ρ	Test acc.	Test AUC
GCN	0.920	0.096	0.90	1.00
AttentiveFP	0.821	0.643	0.60	0.51
MPNN	0.807	-0.104	0.85	0.91
GINE	0.742	0.073	1.00	1.00
GAT	0.718	0.527	0.90	0.99

fore does not select for attribution reliability, which is an argument for auditing the two separately.

5.4 Cross-task patterns

Figure 4 maps both metrics over the full grid and Figure 5 summarises faithfulness by task family. Faithfulness is strongly dataset-dependent rather than a property of the attributor: on the molecular classification sets, mean occlusion ρ is $+0.05$ over 47 cells with 21 of them negative, and the per-dataset means range from clearly negative on BBBP to clearly positive on hERG and SIDER. The same attributor can be faithful on one endpoint and anti-faithful on another.

On exact ground truth the correctness picture is cleaner and instructive (Table 5): with $n = 120$ molecules, the four gradient methods all exceed 0.98 AUROC, GNNExplainer sits at chance (0.501) and PGExplainer falls below it (0.269). The perturbation-based explainers, which are the ones that optimise an explicit mask, are the ones that fail to recover a motif the model provably uses — the model reaches 100% test accuracy on this task.

5.5 Negative and degenerate results

We keep failures in the record.

A metric confound on regression. 16 of 17 regression cells have negative occlusion ρ (mean -0.51). The tempting headline — "attributions are anti-faithful on regression" — is not one we can support, and we flag the reason. Attributions are unsigned magnitudes, whereas Δ_k for a regression head is a *signed* change in the predicted value: removing an important substructure can move a solubility prediction in either direction. Correlating

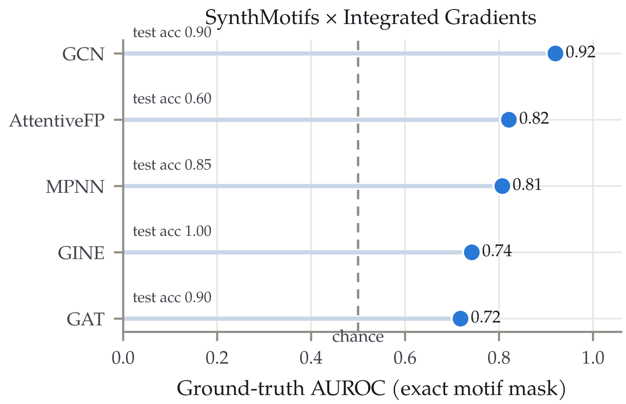

Figure 3: Localisation is backbone-dependent. Ground-truth AUROC on SynthMotifs (exact node labels) with Integrated Gradients held fixed. Test accuracy is annotated: the perfectly accurate backbone (GINE) is not the best explained.

Table 5: Exact-ground-truth control (SynthMotifsXL × GINE, random split, 120 evaluation molecules). The gradient family recovers the injected motif almost perfectly; the two mask-optimising explainers do not, despite the model classifying this task at 100% accuracy.

Attributor	GT AUROC	GT AUPRC	Occ. ρ	Fid+
Guided BP	1.000	0.998	0.529	0.495
Saliency	0.990	0.962	0.424	0.492
Input×Grad.	0.989	0.958	0.393	0.493
Integrated Grad.	0.987	0.945	0.403	0.493
GNNExplainer	0.501	0.259	0.033	0.364
PGExplainer	0.269	0.165	-0.227	0.050

an unsigned quantity with a signed one is not a well-posed faithfulness test, so these numbers are best read as evidence that the occlusion protocol needs a signed or absolute-value variant for regression, not as a property of the attributors. We report the cells because they are part of the run, and mark the interpretation as unresolved.

Degenerate cells. Imbalanced scaffold folds can collapse to a majority-class predictor. BACE × GCN reaches test accuracy 0.133 with AUC 0.224 — worse than chance — and BACE × GINE 0.433 accuracy at AUC 0.434; attributions of a model that has not learned the task are not meaningful, and we exclude such cells from the interpretive claims above while leaving them in the released tables. Conversely, high accuracy can be an artefact of imbalance: Tox21 × GINE shows accuracy 1.000 at AUC 0.827 on a $\sim$4%-positive endpoint, which is why we report AUC alongside accuracy everywhere.

Missing entries. Correctness is undefined for the ten datasets with no per-atom ground truth, and several attributor–dataset combinations were not run. Both appear as --. We do not impute, and we do not report a grid-wide mean over a metric that exists for only 29 of 81 cells.

6 Discussion

What the audit buys. The practical consequence of §5.1–§5.2 is that a faithfulness score should not be reported as if it certified an explanation. In distribution it behaves like a reasonable proxy for correctness; under the scaffold shift that defines realistic MPP evaluation, it stops predicting correctness and can invert. Since ground truth is exactly what is unavailable in the deployment regime, the usable recommendation is not "measure correctness instead" but "calibrate your faithfulness metric on a task where ground truth *is* available, in the same shift regime, before trusting it" — which is what the SynthMotifs control is for.

How attributions should be reported. Our results argue for three changes in practice. First, report attributor *and* backbone: the spread across backbones (0.202 AUROC) is comparable to the spread across attributors, so an explanation is a property of the pair. Second, report the split regime, and report an in-distribution reference alongside the shifted one, since the same metric can behave oppositely in the two. Third, report negative controls: an attributor scoring 0.026 against a chemically motivated mask is making a falsifiable claim, and that is far more informative than a fidelity number with no reference point.

Why reliability is regime-dependent. A faithfulness metric asks what the model uses; a ground-truth metric asks what the chemistry says. The two coincide only when the model has actually learned the causal substructure. In distribution on a synthetic task with an injected motif, that condition holds — the model reaches 100% accuracy using the motif — and faithfulness and correctness agree. Under scaffold shift on real chemistry the condition need not hold: a model can rely on scaffold-correlated shortcuts that generalise poorly, and a faithful attribution will then faithfully report the shortcut. This also explains why the perturbation-based explainers, which optimise a mask against the model's own output, can be at or below chance on a task the model solves: they are optimising fidelity to the model, which is not the same objective as recovering the motif.

7 Limitations and threats to validity

Reduced budget. Every number here comes from CPU-tractable configurations: 30–60 training epochs, 15–25 integration steps, and 20–30 audited molecules for most cells (120 for the exact-ground-truth control). Per-cell estimates are correspondingly noisy, and the bootstrap intervals we compute are wide. The pipeline supports a larger configuration; we do not report it, and no claim here should be read as a converged measurement.

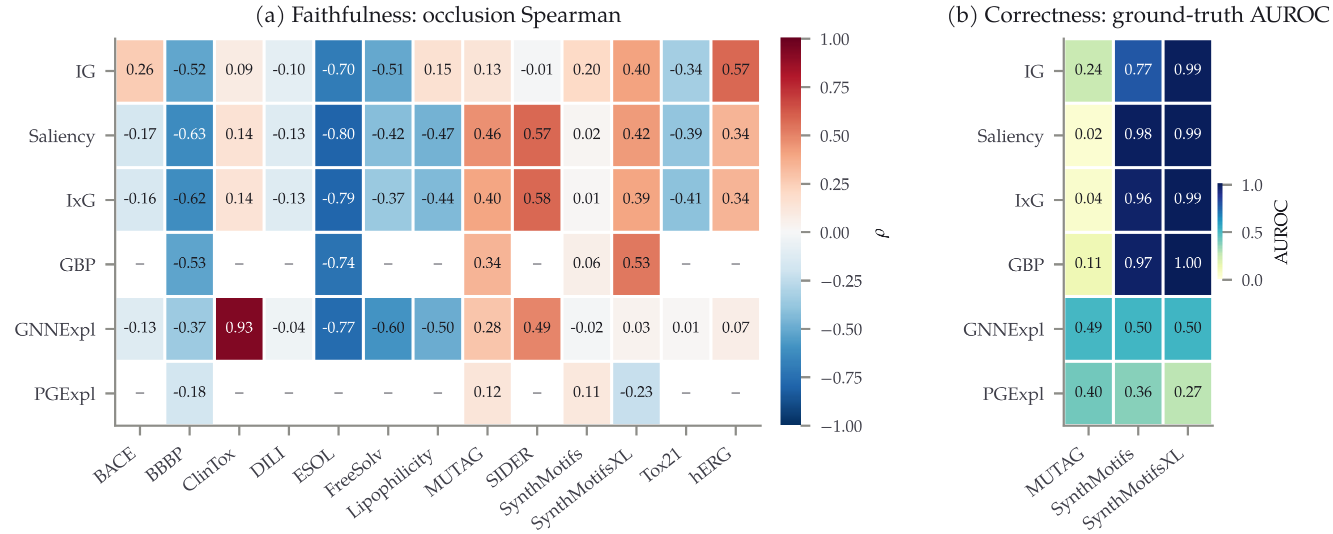

Figure 4: The audit matrix. (a) Occlusion faithfulness across all 13 datasets, averaged over backbones and splits within a cell; blue is anti-faithful, red faithful. (b) Ground-truth correctness, defined only for the three Tier-1 datasets. "–" marks combinations that were not run or where the metric is undefined; nothing is imputed.

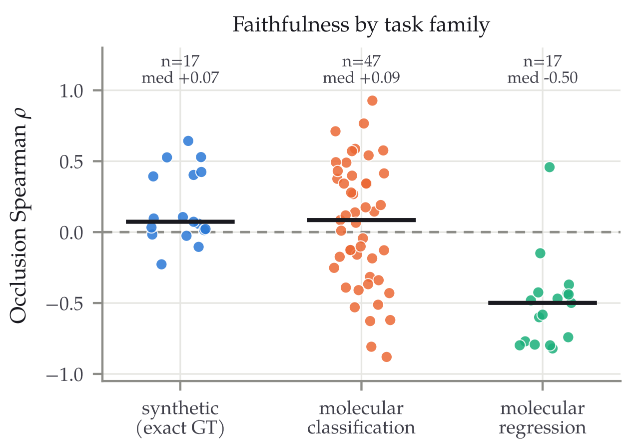

Figure 5: Faithfulness by task family. Each point is one cell. The regression cells sit almost entirely below zero, but see §5.5: for regression this comparison is confounded by attribution sign and should not be read as evidence that regression attributions are anti-faithful.

Single split, single seed. Each cell is one scaffold split at one seed. We therefore make claims about *patterns across many cells* rather than about individual cell values, and the ordering of two adjacent backbones (GINE 0.742 vs. GAT 0.718) should not be over-read.

Ground truth is scarce and partly a proxy. Correctness is measurable on three of 13 datasets. MUTAG's mask is a chemically motivated nitro proxy, not annotator labels: it encodes a hypothesis about mutagenicity, and an attributor could in principle be right about a model that is right for a different reason. The exact-ground-truth results come from synthetic graphs, which are not chemistry. The shift arm of our headline analysis rests on MUTAG at $n = 20$ molecules; it is significant as a paired test, but it is one dataset.

Underpowered rank correlations. The selector correlations in Table 3 are computed over six attributors. Four of six do not reach $p < 0.05$, and we lean on the paired per-molecule tests and on the consistency of the sign pattern instead.

Regression faithfulness is confounded. As set out in §5.5, the unsigned-attribution/signed-Δ mismatch makes the regression ρ_{occ} values uninterpretable as faithfulness. This is a defect in the protocol, not a finding.

Scope. SubgraphX [31] and the GraphXAI [2] synthetic generator could not be installed in our environment (they depend on a pre-2.5 PyTorch Geometric extension stack with no wheel for the installed PyTorch), so both are absent rather than negative results. Our comparison to GraphFramEx is a faithful reimplementation of its characterisation score within our pipeline, not a run of the authors' code.

8 Conclusion

MolSanity audits attribution reliability for molecular GNNs rather than proposing another attributor. Over 81 cells we find that faithfulness and ground-truth correctness are uncorrelated in aggregate ($\rho = -0.03$), that the relationship between them is positive in-distribution and inverts under scaffold shift, that field-standard fidelity metrics consequently mis-select attributors in the shifted regime with per-molecule $p < 0.001$, and that reliability depends on the backbone about as much as on the attributor while being unrelated to predictive accuracy. The practical implication is narrow and, we think, well supported: a faithfulness score is not a certificate of correctness, and should be reported with the backbone, the split regime and a ground-truth reference point attached.

Data and code availability

All code, configurations, per-molecule audit records, figures and the 35 pretrained checkpoints behind every table are in the project repository, https://github.com/Kar488/molsanity. Every number in this paper is generated from the committed `RESULTS.md` and `BENCHMARK_GT.json` by `paper/figs/make_figs.py` and `paper/figs/make_tables.py`; none is transcribed by hand. Because the checkpoints are committed, rerunning the pipeline reproduces the tables without retraining.

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